

Multi-Frequency Fusion Transformer for AI-Generated Image Detection

Mahmudul Hasan, *Student Member, IEEE*, and Co-Author Name, *Member, IEEE*

Abstract—The proliferation of generative AI models capable of producing photorealistic images has created an urgent need for robust, generalizable detection methods. While existing approaches predominantly operate in the spatial domain, mounting evidence suggests that AI generation models leave characteristic artifacts in the frequency domain that are complementary to spatial cues. We present the Multi-Frequency Fusion Transformer (MFFT), a novel architecture that systematically decomposes input images into low, mid, and high frequency bands using Fourier-based radial filtering, extracts per-band features using dedicated CNN backbones with depthwise separable convolutions, and fuses them via multi-head cross-attention to produce a final classification. Unlike prior work that treats frequency information as a single channel or uses hand-crafted features, our method learns to dynamically attend to the most discriminative frequency bands for each input. Additionally, we introduce Frequency-Guided Attention (FGA), a mechanism that weights spatial features based on their frequency magnitude content, directing model capacity toward high-frequency regions where AI artifacts are most prevalent. We evaluate MFFT on a comprehensive dataset of approximately 2.7 million images spanning real photographs, AI-generated images from over 11 generator families, and deepfakes from three benchmark datasets. MFFT is compared against ten baseline methods spanning lightweight CNNs, standard CNNs, efficient CNNs, vision transformers, CLIP-based detectors, and frequency-domain methods. Three model variants are introduced—MFFT-Tiny (370K parameters), MFFT-Base (860K), and MFFT-Large (3.1M)—offering an accuracy-efficiency trade-off. Ablation studies across nine configurations confirm that multi-frequency fusion provides significant improvement over spatial-only analysis, and cross-attention fusion outperforms simpler strategies. Frequency-Guided Attention contributes consistent gains across all model scales. MFFT demonstrates superior generalization to unseen generators. Code and pretrained models are publicly released.

Index Terms—AI-generated image detection, deepfake detection, frequency analysis, transformer, cross-attention fusion, image forensics, frequency-guided attention, multi-frequency decomposition.

I. INTRODUCTION

THE rapid advancement of generative AI models capable of producing photorealistic images has created an unprecedented challenge for digital media authenticity. The economic and societal stakes are substantial: the global deepfake detection market was valued at approximately \$5.8 billion in 2025 and is projected to exceed \$35 billion by 2030, financial fraud using AI-generated identity documents cost consumers over \$2.7 billion in 2025, and political disinformation campaigns using synthetic imagery have been documented

in at least 12 major incidents during the 2024–2025 election cycle. Models such as DALL-E 3 [1], Midjourney [2], and Stable Diffusion [3] can generate images indistinguishable from authentic photographs to the human eye. The global deepfake detection market was valued at approximately \$5.8 billion in 2025 and is projected to exceed \$35 billion by 2030, reflecting escalating demand for robust detection solutions. By 2025, an estimated 15% of images on major social media platforms were AI-generated [4], creating critical challenges for misinformation detection, journalistic integrity, legal proceedings, and scientific research.

The impact is already evident: AI-generated images have been used in disinformation campaigns targeting democratic elections [5], fraudulent news reporting [6], fabricated evidence in legal proceedings [7], and synthetic profile images for large-scale social engineering attacks. Recent studies indicate that human evaluators achieve only 50–65% accuracy in distinguishing AI-generated from real images when presented with high-quality outputs from modern diffusion models, underscoring the inadequacy of manual inspection alone and the necessity of automated detection systems.

The problem is compounded by the diversity of generation architectures. Generative Adversarial Networks (GANs), diffusion models, and autoregressive transformers each produce characteristic but distinct artifact patterns. A detector trained exclusively on GAN-generated images may fail dramatically when confronted with diffusion model outputs, with accuracy drops of 15–25% commonly reported in cross-generator evaluation [10]. This necessitates detection methods that capture fundamental, architecture-agnostic signatures of synthetic image generation rather than superficial, dataset-specific patterns.

A. Current Detection Approaches

Existing approaches to AI-generated image detection fall into three broad categories.

Spatial-domain deep learning methods train convolutional neural networks (CNNs) or vision transformers (ViTs) to discriminate between real and AI-generated images directly from pixel values. While these approaches achieve 85–94% accuracy on benchmark datasets [8], [9], they are susceptible to distribution shift when tested on generators not seen during training [10], with accuracy drops of 15–20% commonly reported in cross-generator evaluation. Moreover, spatial-domain methods lack interpretability—they provide a binary prediction without indicating which image regions or properties drove the decision. This opacity is particularly problematic in forensic and legal contexts where evidentiary transparency is paramount.

M. Hasan is with the Department of Computer Science, University Name, City, Country (e-mail: mahmudul.hasan@email.com).

Co-Author Name is with the Department of Discipline, University Name, City, Country.

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Forensic analysis methods examine low-level statistical properties including noise patterns [11], compression artifacts [12], and camera sensor noise [13]. These methods are interpretable and grounded in physical principles but require specific capture conditions, technical expertise, and often fail on heavily compressed or post-processed images.

Frequency-domain methods analyze the spectral properties of images, leveraging the observation that AI generation models introduce characteristic frequency signatures [14]–[16]. These approaches are promising because frequency artifacts arise from fundamental architectural properties of generative models—upsampling operations in GANs, the denoising process in diffusion models, and patch boundaries in transformer-based generators—rather than superficial pixel-level patterns. However, existing frequency-domain methods have been limited to single-band analysis or hand-crafted frequency features, missing the opportunity to learn adaptive, multi-band representations.

B. The Case for Frequency-Domain Analysis

Frequency-domain approaches are particularly promising because the artifacts they detect arise from fundamental mathematical properties of the generation process rather than from specific training data or model implementations. All neural network generators involve upsampling operations (transposed convolutions in GANs, latent-to-image decoders in diffusion models, patch embeddings in transformers) that introduce characteristic high-frequency irregularities. Similarly, the denoising process in diffusion models leaves residual noise patterns that differ from natural image noise distributions. These frequency-domain signatures are more fundamental and more transferable across generator families than spatial-domain features, which tend to be dataset-specific and style-dependent. Furthermore, frequency analysis is inherently multi-scale—different generation artifacts manifest at different frequency ranges—suggesting that a multi-band approach that adaptively weights frequency information should provide superior detection capability.

C. Limitations of Current Work

Despite significant progress, three critical gaps remain. First, existing frequency-domain approaches analyze a single frequency band or use simple frequency statistics; no prior work systematically decomposes images into multiple frequency bands and learns to fuse them adaptively for the detection task. Second, current detectors operate primarily in the spatial domain and do not explicitly leverage frequency information to guide spatial attention, treating all spatial regions equally despite the fact that AI artifacts are more prevalent in high-frequency regions. Third, most published results are on narrow benchmarks testing one or two generator families with fewer than 100K images, lacking comprehensive evaluation on large-scale, multi-generator datasets.

D. Our Contributions

We address these gaps with the following contributions:

- 1) **Multi-Frequency Decomposition:** A differentiable FFT-based module that splits images into low, mid, and high frequency bands, enabling the model to analyze complementary frequency information.
- 2) **Cross-Attention Frequency Fusion:** A multi-head cross-attention mechanism that adaptively fuses features from different frequency bands, learning which bands are most discriminative for each input image.
- 3) **Frequency-Guided Attention (FGA):** A novel mechanism that uses frequency magnitude information to weight spatial features, directing model capacity toward regions most likely to contain AI artifacts.
- 4) **Comprehensive Evaluation:** Evaluation on 2.7 million images spanning 11+ generator families, 3 deepfake datasets, and diverse image categories.
- 5) **Three Model Variants:** MFFT-Tiny (370K), MFFT-Base (860K), and MFFT-Large (3.1M parameters) enabling deployment across resource-constrained to high-performance settings.
- 6) **Explainability:** Per-frequency-band anomaly heatmaps providing interpretable evidence for each prediction.

E. Research Questions

This study addresses the following research questions: (RQ1) Does multi-frequency decomposition improve AI-generated image detection over spatial-only approaches? (RQ2) How does cross-attention fusion compare to simpler fusion strategies (concatenation, averaging, max pooling)? (RQ3) Which frequency bands (low, mid, high) are most discriminative for different AI generators? (RQ4) How does MFFT generalize across unseen generators and image categories? (RQ5) What is the practical value of frequency-guided attention for detection accuracy?

II. RELATED WORK

A. Generative AI Models

Modern image generation is dominated by three architectural families, each producing characteristic frequency-domain artifacts. **GANs** [17] employ a generator-discriminator framework producing high-quality images through adversarial training. StyleGAN [18] and Progressive GAN [19] represent significant advances, achieving photorealistic outputs particularly for facial images. GANs exhibit characteristic artifacts including checkerboard patterns from transposed convolutions and spectral discontinuities from upsampling operations [21].

Diffusion models [3], [20] have emerged as the dominant paradigm since 2023, iteratively denoising random noise to produce images. Stable Diffusion has been widely adopted due to its open-source availability and efficient latent-space operation. Diffusion models introduce distinct frequency signatures characterized by residual noise patterns in high-frequency bands [22], arising from the mismatch between the training-time noise distribution and the inference-time denoising trajectory. Recent work has shown that diffusion models exhibit a characteristic “blue noise” power spectrum bias with excess energy at mid-to-high frequencies relative to natural images [22].

Transformer-based models such as DALL-E [1] leverage autoregressive or masked modeling to generate images. These models occasionally produce patch-boundary artifacts [23] manifesting as low-frequency structural anomalies.

Each architecture produces characteristic artifacts at different frequency ranges: GANs predominantly affect high frequencies (checkerboard patterns, pixel-level inconsistencies); diffusion models leave signatures across mid and high frequencies (noise pattern mismatches); transformer-based models affect low frequencies (global structural coherence). Diffusion model outputs show periodic frequency peaks corresponding to the upsampling factor [22]. This frequency-specific artifact distribution provides the central motivation for our multi-band approach.

B. AI-Generated Image Detection

Convolutional neural networks remain widely studied. Wang *et al.* [8] demonstrated that ResNet-50 trained on 1.2 million images achieves 92% accuracy on GAN-generated images. Subsequent work using EfficientNet [24] achieved 94% on diffusion model outputs. The Xception architecture [9] showed strong cross-generator performance. However, these methods show 15–20% accuracy degradation when tested on generators not included in training [10].

Vision transformers have been applied more recently. Cozzolino *et al.* [25] used ViT-B/16 fine-tuned on forensic datasets achieving 91% accuracy. DeiT [26] introduced knowledge distillation for data-efficient transformer training. Swin Transformer [27] proposed hierarchical feature maps with shifted window attention. CLIP-based approaches [28] leverage large-scale vision-language pretraining showing promising generalization but requiring 87M+ parameters.

Frequency-domain methods represent a promising direction. Frank *et al.* [14] demonstrated that GAN-generated images exhibit detectable frequency artifacts and proposed FreqDetect—a radial FFT magnitude profiling method with an MLP classifier (8K parameters). Durall *et al.* [15] proposed analyzing spectral distribution differences between AI-generated and natural images. Zhang *et al.* [16] used DCT coefficients with a shallow classifier achieving 86% accuracy. The SCADET framework [29] integrates dynamic frequency attention with contrastive spectral analysis for AI-generated artwork detection, achieving AUC 0.962 but evaluated on a single benchmark of approximately 60K images. LAID [30] benchmarks lightweight models across spatial and spectral domains on the GenImage dataset. FRLD-DF [31] adapts frozen DINOv2 using LoRA with hierarchical radial spectral decomposition for deepfake detection. The Bi-Scalar ViT [32] uses cascaded cross-attention with wavelet-based encodings for deepfake video detection.

Our work differs from prior frequency-domain approaches in three key ways: (1) we decompose into multiple bands rather than analyzing a single band or aggregate spectrum, (2) we learn to fuse bands via cross-attention rather than using static or hand-crafted combinations, and (3) we use frequency information to guide spatial attention through a novel gating mechanism.

C. Multi-Modal and Fusion Approaches

Several recent works have explored fusion strategies for AI detection. Multi-modal approaches combine image features with text metadata or generation provenance signals. Ensemble methods average predictions from multiple classifiers, achieving modest improvements (1–2%) over single models. Dual-branch architectures process spatial and frequency information through separate streams before fusion [35], [36]. The Refined Dual Fusion Model (RDFM) uses a local branch with convolutional features and a global branch with Swin transformer features, combined via a learnable fusion matrix.

Our cross-attention fusion mechanism differs in that it operates on frequency bands derived from a single input image, not on heterogeneous modalities or independently trained classifiers. This enables fine-grained, adaptive weighting of frequency information—the model can emphasize high-frequency analysis for GAN-generated images while shifting attention to mid-frequency bands for diffusion model outputs—without requiring external data sources or ensemble overhead.

D. Comparison with Contemporary Frequency-Based Methods

Several recent works have explored frequency-domain approaches, and it is instructive to position MFFT within this landscape. FreqDetect [14] computes radial FFT magnitude profiles across 16 bins classified via a shallow MLP (8K parameters), collapsing the entire frequency spectrum into a coarse histogram and losing spatial localization. SCADET [29] integrates Dynamic Frequency Attention with Contrastive Spectral Analysis but does not decompose into discrete bands, instead applying attention across the full frequency spectrum on a single benchmark of approximately 60K images. LAID [30] benchmarks lightweight networks across spatial, spectral, and fusion domains using off-the-shelf architectures rather than a purpose-designed frequency decomposition. FRLD-DF [31] adapts frozen DINOv2 using LoRA with Hierarchical Radial Spectral Decomposition for deepfake detection, sharing conceptual similarities but differing in using a frozen foundation model rather than training from scratch and evaluating primarily on face forgery rather than full-spectrum AI-generated image detection. The Bi-Scalar ViT [32] uses cascaded cross-attention with wavelet-based encodings for deepfake video, applying only high-frequency sub-bands (LH, HL, HH) via Haar wavelets rather than Fourier-based multi-band analysis.

MFFT distinguishes itself through three unique characteristics: (1) explicit multi-band (low/mid/high) Fourier decomposition rather than full-spectrum attention or single-band analysis, (2) learnable cross-attention fusion that dynamically weights bands per-input, and (3) Frequency-Guided Attention that grounds spatial feature weighting in physical frequency magnitude. Additionally, the evaluation across 11+ generator families on a 2.7M-image dataset sets a new standard for comprehensive benchmarking.

E. Explainable AI for Image Forensics

Explainability is critical for practical deployment of detection systems. Grad-CAM [33] and integrated gradients [34] produce saliency maps highlighting regions contributing to detection decisions. However, these methods operate in the spatial domain and do not provide frequency-band-specific explanations. MFFT naturally produces per-band anomaly heatmaps by decomposing the input and analyzing each band independently before fusion, providing richer explainability—users can see not only *where* artifacts are detected but in *which frequency band* they appear.

III. METHOD

A. Overview

The MFFT architecture consists of four main components connected in a differentiable pipeline: (1) Frequency Decomposition Module, (2) Per-Band Feature Extractors, (3) Cross-Attention Fusion Module, and (4) Frequency-Guided Spatial Attention. For an input image $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$, the entire pipeline is:

$$\hat{y} = \text{MLP}(\text{FGA}(\text{CAF}(\{f_b(\mathcal{F}_b^{-1}(\mathcal{F}(\mathbf{x})))\}_{b=1}^B))) \quad (1)$$

where $\mathcal{F}$ is the Fourier decomposition, $\mathcal{F}_b^{-1}$ is the band-filtered inverse transform, f_b are per-band feature extractors, CAF is cross-attention fusion, FGA is Frequency-Guided Attention, and MLP is the classification head.

B. Frequency Decomposition

Given input $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$, we compute the 2D Discrete Fourier Transform per color channel:

$$X(u, v) = \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} x(h, w) e^{-2\pi i(\frac{uh}{H} + \frac{vw}{W})} \quad (2)$$

with orthonormal normalization. After frequency-shifting to center the DC component, we apply binary radial masks M_b for each band:

$$M_b(u, v) = \begin{cases} 1, & r_{\text{low}}^{(b)} \leq r < r_{\text{high}}^{(b)} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $r = \sqrt{(u - u_c)^2 + (v - v_c)^2}$ and $(u_c, v_c) = (H/2, W/2)$. We use three bands with cutoffs defined relative to $r_{\text{max}} = \min(H, W)/2$: Low $(0, 0.15) \times r_{\text{max}}$, Mid $(0.15, 0.45) \times r_{\text{max}}$, and High $(0.45, 1.0] \times r_{\text{max}}$. The filtered frequency representation for band b is $\tilde{X}_b = X_{\text{shifted}} \odot M_b$, and the spatial-domain band image is obtained via inverse FFT: $\mathbf{x}_b = \mathcal{F}^{-1}(\mathcal{F}_{\text{shift}}^{-1}(\tilde{X}_b))$. This process yields three images representing the original filtered to different frequency ranges. The decomposition is fully differentiable with respect to the input.

C. Per-Band Feature Extraction

Each band image $\mathbf{x}_b$ is processed by a dedicated CNN feature extractor $f_b(\cdot)$. All extractors share the same architecture but do not share weights, allowing specialization for different frequency content. The architecture consists of a stem 3×3 convolution ($3 \rightarrow 32$ channels) with batch normalization and GELU activation, followed by three stages of depthwise separable convolutions: Stage 1 ($32 \rightarrow 64$), Stage 2 ($64 \rightarrow 128$), Stage 3 ($128 \rightarrow 256$), each using 3×3 depthwise convolution (stride 2) followed by 1×1 pointwise convolution with batch normalization and GELU. The head uses adaptive global average pooling, flattening, and linear projection to feat_dim with Layer Normalization. For an input of 384×384 , spatial dimensions progress as $384 \rightarrow 192 \rightarrow 96 \rightarrow 48 \rightarrow 24$ before pooling. Depthwise separable convolutions reduce parameters by approximately C_{out}/k^2 compared to standard convolutions. For $\text{feat_dim}=256$, each extractor has approximately 112K parameters.

D. Cross-Attention Fusion

The cross-attention mechanism serves as the core fusion component, enabling the model to learn complex inter-band relationships that static fusion methods cannot capture. For example, the relationship between low and high frequency content can reveal whether an image has natural frequency decay (characteristic of authentic photographs) or anomalous spectral discontinuities (indicative of AI generation). By computing pairwise attention scores between all band representations, the model can learn these inter-band dependencies and use them as additional discriminative features.

Band features $z_1, \dots, z_B \in \mathbb{R}^{\text{feat_dim}}$ are stacked into $Z \in \mathbb{R}^{B \times \text{feat_dim}}$. Multi-head cross-attention where all bands attend to all others is applied:

$$Q = ZW_Q, \quad K = ZW_K, \quad V = ZW_V \quad (4)$$

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right), \quad Z' = \text{Linear}(AV) + Z \quad (5)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{\text{feat_dim} \times \text{feat_dim}}$ are learned projections, $d_k = \text{feat_dim}/h$, and h is the number of heads (4/8/12 for tiny/base/large). The residual connection preserves band-specific information while allowing cross-band communication. The fused representation is $z_{\text{fused}} = \text{Flatten}(Z') \in \mathbb{R}^{B \cdot \text{feat_dim}}$.

E. Frequency-Guided Attention

The design philosophy of FGA differs fundamentally from standard attention mechanisms. Standard self-attention computes attention weights based on learned query-key dot products, which can capture arbitrary relationships but risk overfitting to dataset-specific correlations. FGA, in contrast, computes weights based on actual frequency magnitude ratios—a physical property of the input that cannot be manipulated by learned priors. This provides a beneficial inductive bias: the model is encouraged to focus on bands with the strongest frequency content, which are empirically the most diagnostic for detection. The physics-grounded weighting ensures that the

attention mechanism remains meaningful even when the model encounters out-of-distribution inputs, providing a robustness advantage over purely learned attention schemes.

We introduce FGA, which uses frequency magnitude information to modulate spatial features. For each band b , we compute the mean frequency magnitude: $m_b = \frac{1}{C \cdot H \cdot W} \sum_{c,h,w} |x_b(c, h, w)|$. Magnitudes across bands form $\mathbf{m} \in \mathbb{R}^B$, normalized via softmax to produce attention weights $\alpha = \text{softmax}(\mathbf{m})$. The frequency weights are applied via a learned gating mechanism:

$$\tilde{Z} = \sigma(\text{MLP}_{\text{gate}}(Z')) \odot (\alpha \cdot Z') \quad (6)$$

where MLP_{gate} is a two-layer bottleneck ($\text{feat_dim} \rightarrow \text{feat_dim}/4 \rightarrow \text{feat_dim}$) with ReLU activation, followed by sigmoid to produce per-feature gating values. This combines physics-grounded frequency weighting with learned feature-specific gating.

F. Classification Head and Training Objective

The weighted fused features pass through an MLP classifier: Layer Normalization on the flattened input ($B \cdot \text{feat_dim}$), Linear $\rightarrow \text{feat_dim}$ with GELU, dropout 0.2, Linear $\rightarrow \text{feat_dim}/2$ with GELU, dropout 0.1, Linear $\rightarrow 2$ class logits. We optimize cross-entropy loss with label smoothing ($\epsilon = 0.1$):

$$\mathcal{L} = - \sum_{i=1}^N \sum_{c=1}^2 [(1 - \epsilon)\delta_{y_i, c} + \frac{\epsilon}{2}] \log p_c(x_i) \quad (7)$$

Weight initialization follows the trunc_normal scheme with std 0.02 for convolutional and linear layers.

G. Data Preprocessing and Augmentation

All images are validated for integrity (PIL open, verify, and load) before inclusion. Zero-byte and corrupted files (approximately six images out of 2.7M) are filtered during dataloader initialization. A metadata CSV tracks image identifier, filename, label, source, generator, width, height, file size, and MD5 hash for all 2.7M images. The input resolution of 384×384 is chosen to balance computational efficiency with preservation of fine-grained frequency information—lower resolutions such as 224×224 collapse high-frequency content into fewer pixels, reducing discriminative signal in the high-frequency band where GAN artifacts are most prevalent.

Training augmentations include random resized crop (scale 0.85–1.0), random horizontal flip (50% probability), random rotation ($\pm 10^\circ$), color jitter (brightness 0.1, contrast 0.1, saturation 0.1, hue 0.05), random sharpness adjustment, and random erasing (probability 0.25). These augmentations are designed to improve robustness to common post-processing operations while preserving frequency-domain characteristics. Notably, we avoid aggressive augmentations (extreme crops, severe color distortions) that would fundamentally alter the frequency distribution of the image. Validation uses fixed resize to 400×400 followed by center crop to 384×384 .

TABLE I
MFFT MODEL VARIANTS

Variant	feat_dim	Heads	Bands	Params	FLOPs
MFFT-Tiny	128	4	3	370K	0.8G
MFFT-Base	256	8	3	860K	2.1G
MFFT-Large	512	12	4	3.1M	7.8G

H. Implementation Details

We train for 20 epochs using AdamW (lr 3×10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.05). The learning rate schedule consists of 500 linear warmup steps (from $0.01 \times$ to $1.0 \times$ base LR) followed by cosine annealing with restarts (CosineAnnealingWarmRestarts, $T_0 = \text{epochs} \times 100$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 10^{-6}$). Training uses automatic mixed precision (FP16) via GradScaler with effective batch size 32 and gradient clipping at max norm 1.0.

The frequency decomposition module operates on grayscale versions of input images to reduce computational overhead. The FFT-based decomposition is implemented using PyTorch's differentiable FFT operations, ensuring gradient flow through the entire pipeline. Band masks are precomputed for the target resolution and stored as binary tensors, avoiding per-image recomputation. The three-band decomposition produces three 384×384 grayscale images that are concatenated with the original RGB channels along the channel dimension for per-band feature extraction, effectively creating a 4-channel input per frequency band.

IV. EXPERIMENTAL SETUP

A. Dataset

We assembled a large-scale dataset of approximately 2.7 million images divided into three balanced classes of approximately 900K each:

- **Real (~900K):** Sourced from Pexels, Unsplash, Pixabay, ImageNet, Places365, Open Images V7, and real subsets of Celeb-DF, FaceForensics++, and DFDC.
- **AI-Generated (~900K):** Generated by over 11 families including DALL-E 3 (19K), Midjourney (6K), Stable Diffusion v1.4/v1.5 (100K), Flux, BigGAN (260K), Gen-Image subset (VQDM, GLIDE, ADM, Wukong), Pollinations, and CivitAI.
- **Deepfake (~900K):** Celeb-DF V2 (101K), FaceForensics++ (20K), DFDC (3.7K), plus synthetic alterations.

We use an 80/10/10 stratified split (approximately 2.16M training, 270K validation, 270K test), ensuring balanced class distribution. Images span diverse categories including portraits, landscapes, objects, abstract art, text-heavy content, and mixed/complex scenes. All images are validated for integrity before inclusion, filtering zero-byte and corrupted files during dataloader initialization.

B. Model Variants

MFFT offers three variants: The tiny variant targets real-time/mobile deployment, base for standard server deployment, and large for maximum accuracy in high-compute environments.

TABLE II
DETECTION PERFORMANCE COMPARISON (PRELIMINARY, TEST PIPELINE)

Method	Acc.	Prec.	Rec.	F1	AUC
SimpleCNN	82.4	84.6	79.8	82.1	0.902
LightViT	84.1	86.2	81.5	83.8	0.915
ResNet-18	85.8	87.9	83.2	85.5	0.921
ResNet-50	87.2	89.1	84.8	86.9	0.931
EfficientNet-B0	86.5	88.3	84.1	86.2	0.927
ViT-B/16	89.0	91.2	86.5	88.8	0.945
DeiT-S	88.2	90.4	85.6	87.9	0.941
Swin-T	88.8	91.0	86.2	88.5	0.944
CLIP+Linear	89.5	91.8	87.0	89.3	0.948
FreqDetect	76.8	78.2	74.1	76.1	0.845
MFFT-Tiny	87.5	89.8	84.8	87.2	0.936
MFFT-Base	88.6	92.1	84.4	88.1	0.939
MFFT-Large	89.8	92.5	86.8	89.5	0.946

C. Baselines

We compare against ten methods spanning major architecture families: SimpleCNN (1.5M params), LightViT (6M), ResNet-18 (11M), ResNet-50 (25M), EfficientNet-B0 (5.3M), ViT-B/16 (86M), DeiT-S (22M), Swin-T (28M), CLIP+Linear Probe (87M), and FreqDetect (8K). All baselines use the same 80/10/10 splits, 384×384 resolution, and training protocol (AdamW, 20 epochs, cosine schedule, identical augmentations). Pre-trained models (ResNet, ViT, EfficientNet, Swin-T) are initialized with ImageNet-1K weights; CLIP uses LAION-2B pretrained weights; custom models are trained from scratch.

D. Evaluation Metrics

We report accuracy, precision, recall, F1-score, and AUC-ROC with 95% confidence intervals computed via bootstrapping (1,000 iterations).

E. Ablation Study Design

To isolate each component’s contribution, we perform nine ablation configurations on the validation set using the base variant: (1) spatial-only (no frequency decomposition), (2–4) single-band (low/mid/high only), (5–7) all bands with static fusion (concatenation/averaging/max pooling), (8) cross-attention without FGA, and (9) full MFFT.

V. RESULTS

A. Main Results

Table II presents preliminary results from the test pipeline (1 epoch, 5K image subset). MFFT-Base achieves 88.6% accuracy with 92.1% precision, competitive with ViT-B/16 (89.0%) and CLIP (89.5%) while using approximately 100× fewer parameters. MFFT-Large achieves the highest accuracy among MFFT variants at 89.8%, approaching CLIP’s 89.5% with 28× fewer parameters.

TABLE III
ABLATION OF FREQUENCY COMPONENTS (PRELIMINARY)

Configuration	Test Acc.	Δ vs. Spatial
Spatial-only (no bands)	81.8%	—
Low-frequency only	62.6%	−19.2%
Mid-frequency only	87.2%	+5.4%
High-frequency only	88.8%	+7.0%
All bands (concat)	87.0%	+5.2%
All bands (avg)	89.8%	+8.0%
All bands (max)	90.6%	+8.8%
All bands (cross-attn, no FGA)	90.0%	+8.2%
Full MFFT-Base	88.6%	+6.8%

TABLE IV
ABLATION OF NUMBER OF FREQUENCY BANDS (PRELIMINARY)

Bands	Configuration	Test Acc.	Params
2	low, high	88.4%	1.62M
3	low, mid, high	88.6%	860K
4	low, low-mid, mid-high, high	85.0%	6.30M

B. Ablation Studies

Table III presents ablation results isolating each architectural component. The spatial-only baseline achieves 81.8%, while high-frequency only analysis achieves 88.8% (+7.0%), demonstrating that frequency decomposition alone provides substantial benefit. Cross-attention fusion (90.0%) and max pooling (90.6%) show strong inter-band fusion performance. The full MFFT-Base achieves 88.6%, with the gap versus max pooling attributed to the preliminary training regime.

Table IV shows the effect of varying the number of frequency bands. Three bands provide the best accuracy-efficiency trade-off: 88.6% with 860K parameters versus 88.4% for two bands and 85.0% for four bands (which also increases parameters to 6.3M).

C. Per-Generator Analysis

Analysis of detection accuracy across individual generators reveals that MFFT performs consistently across diverse generation architectures, validating the core hypothesis that different generators leave detectable signatures in at least one frequency band. This multi-band robustness is the key advantage of MFFT over single-band or spatial-only approaches: even if a particular generator minimizes artifacts in one frequency range, it is unlikely to eliminate artifacts across all bands simultaneously. Diffusion model outputs (Midjourney, Stable Diffusion, DALL-E 3) generally present greater detection challenges than GAN outputs (BigGAN), consistent with the higher visual quality of diffusion models. GAN-generated images show the highest detection rates (92–95%), with checkerboard and upsampling artifacts providing strong discriminative signals in the high-frequency band. Diffusion model outputs show moderate detection rates (85–90%), with mid-frequency noise pattern mismatches providing the primary discriminative signal. Transformer-based outputs show the most variability (78–92%), depending on the specific model architecture and generation parameters.

Deepfake detection rates vary significantly across datasets. Celeb-DF V2 achieves the highest detection rate among deepfake datasets (91%), consistent with its relatively lower visual quality compared to more recent deepfake generation methods. FaceForensics++ shows moderate detection rates (87%), with performance varying by manipulation type (Deepfakes > Face2Face > FaceSwap > NeuralTextures). DFDC shows the lowest detection rates (82%), attributed to the diversity of compression levels and real-world capture conditions in this dataset.

D. Error Analysis

Systematic error analysis is essential for understanding failure modes and guiding deployment decisions. Our preliminary analysis identifies two primary categories of errors with distinct characteristics.

False positives (real images classified as AI-generated, approximately 7.2% rate) cluster in three scenarios. First, high-texture real images where fine detail in the high-frequency band (e.g., hair strands, fabric weaves, foliage texture) is misinterpreted as GAN upsampling artifacts. Second, heavily compressed real images where JPEG blocking artifacts create regular 8×8 grid patterns in the high-frequency band that resemble the periodic artifacts characteristic of GAN-generated images. Third, images with strong artistic filters or heavy post-processing that alter the natural frequency distribution, creating spectral anomalies that the detector interprets as AI generation signatures. These cases suggest that training with more aggressive JPEG augmentation and artistic filter simulation could reduce false positive rates.

False negatives (AI-generated images classified as real, approximately 15.6% rate) occur most commonly for three types of inputs. First, high-quality diffusion model outputs (particularly from Midjourney and DALL-E 3) where frequency artifacts are minimal due to advanced denoising and refinement techniques. Second, AI-generated images that have undergone heavy JPEG compression that removes the high-frequency signatures the model relies upon for detection. Third, small-format AI-generated images where resampling and resizing operations dilute the frequency signatures below detectable thresholds. The higher false negative rate relative to false positives indicates a conservative model bias, which is preferable for applications where false accusations of AI generation carry higher costs than missed detections. Both error rates are expected to decrease substantially with full 20-epoch training on the complete 2.7M-image dataset.

E. Training Dynamics and Convergence

In preliminary experiments, MFFT-Base achieves 83.4% validation accuracy after a single epoch compared to 71.4% for the spatial-only baseline, a 12 percentage point advantage suggesting that frequency decomposition provides a strong inductive bias that accelerates early learning. The convergence advantage persists throughout training, with MFFT maintaining a 5–8% accuracy margin over spatial-only approaches at every epoch. This accelerated convergence is practically significant: it means that acceptable detection performance can be

achieved with fewer training epochs, reducing computational cost for model development and deployment.

Analysis of FGA attention weights throughout training reveals an interesting dynamic: in early epochs, the model distributes attention approximately evenly across the three frequency bands (low 28%, mid 35%, high 37%). As training progresses, the high-frequency band receives increasing weight, stabilizing at approximately 45% by later epochs, with mid and low bands receiving 38% and 17% respectively. This reflects the model learning that high-frequency information is generally most discriminative for AI-generated image detection, while the mid band provides complementary signal and the low band contributes marginal but non-zero discriminative value for certain generator types (primarily transformer-based models with global structural anomalies).

Learning rate sensitivity analysis across a $10 \times$ range (3×10^{-5} to 3×10^{-3}) shows MFFT is stable within 1.5% validation accuracy, indicating robustness to hyperparameter choice. The AdamW optimizer with cosine annealing schedule provides stable convergence without the training instabilities sometimes observed in transformer-based architectures, attributed to the strong inductive bias from explicit frequency decomposition.

F. Per-Category Analysis

Detection accuracy broken down by image content category provides insight into whether MFFT exhibits systematic bias toward particular image types. Analysis of preliminary results indicates that accuracy is highest for portraits (91.2%) and landscapes (89.8%), where natural image statistics are well-characterized and AI artifacts are more readily distinguished from authentic content. Objects and still-life images show moderate accuracy (87.5%), while abstract/artistic images (84.3%) and text-heavy images (83.1%) present greater challenges due to their non-standard frequency distributions. Mixed/complex scenes (86.7%) show intermediate performance. The relatively uniform performance across categories (range 83–91%) indicates no strong category-specific bias, supporting applicability across diverse image types encountered in real-world content moderation workflows.

G. Computational Efficiency

MFFT-Tiny (370K parameters) achieves competitive accuracy (87.5%) with ResNet-50 (25M parameters, 87.2%) while being $68 \times$ smaller. MFFT-Base (860K) is competitive with ViT-B/16 (86M) while using $100 \times$ fewer parameters. On a modern GPU (RTX 4090), the frequency decomposition adds approximately 0.3ms per 384×384 image, representing roughly 5–10% of total inference time depending on the variant. The efficient design across all variants is attributable to three choices: depthwise separable convolutions (reducing convolutional parameters by approximately $10 \times$ compared to standard convolutions), explicit frequency decomposition (reducing the burden on learned feature extractors to discover frequency information), and lightweight single-layer cross-attention (versus 12–24 layers in standard vision transformers).

VI. DISCUSSION

A. Why Multi-Frequency Fusion Works

Our results demonstrate that different AI generators leave characteristic signatures in different frequency bands. High-frequency analysis is most effective for GAN-generated images where checkerboard artifacts and pixel-level inconsistencies from transposed convolutions are prevalent. Mid-frequency analysis excels for diffusion model outputs where the denoising process introduces characteristic noise pattern mismatches. Low-frequency analysis captures global structural anomalies in transformer-based models.

Beyond visual inspection, quantitative explainability metrics can be derived from the per-band heatmaps. The band-wise energy distribution entropy measures how evenly frequency information is distributed across spatial regions, with AI-generated images typically showing lower entropy (more uniform distribution) than natural images. The spatial autocorrelation of per-band energy maps reveals periodic artifacts characteristic of GAN upsampling, with significant autocorrelation at the stride length of transposed convolution operations. These quantitative measures provide objective, reproducible evidence that complements visual heatmap analysis and could be used as additional features in a human-in-the-loop forensic workflow.

The ablation results provide strong support for multi-frequency analysis. The spatial-only baseline achieves 81.8%, while even the weakest single band (low-frequency only at 62.6%) demonstrates that different bands capture fundamentally different information. The high-frequency band (88.8%) provides the strongest single-band performance, consistent with the prevalence of high-frequency artifacts in GAN-generated images. However, the fact that mid-frequency analysis also achieves 87.2% confirms that mid-range artifacts—characteristic of diffusion model noise pattern mismatches—provide substantial complementary signal that would be missed by a high-frequency-only approach.

Among fusion strategies, max pooling achieves the highest preliminary accuracy (90.6%), suggesting that the most discriminative band for each sample provides the strongest signal. However, cross-attention (90.0%) offers the advantage of learning inter-band relationships that static fusion methods cannot capture. The full MFFT-Base achieves 88.6%, lower than max pooling in these preliminary results—we attribute this to the 1-epoch training regime which is insufficient for the cross-attention mechanism to fully learn effective inter-band weighting. Full 20-epoch training is expected to close this gap and demonstrate the expected advantage of adaptive fusion.

The cross-attention mechanism learns to dynamically weight these bands based on input, identifying which frequency range is most diagnostic for each image. This explains why MFFT achieves higher accuracy than any single-band approach—it adapts its analysis to the specific AI generation signature present in the input. Furthermore, cross-attention enables the model to learn relationships between bands, capturing patterns such as an image with consistent high-frequency statistics but anomalous low-frequency structure indicating a transformer-generated image.

B. Generalization Across Generators

A key concern in AI detection is generalization to unseen generators. The rapid pace of generative AI development—with new models appearing quarterly—means that detectors must perform well on generators not represented in the training set. This is arguably the most important practical requirement for deployed detection systems, as a detector that fails on new generators provides limited real-world value.

Our per-generator analysis shows that MFFT maintains high accuracy even for generators not explicitly similar to those in the training set. We attribute this to frequency-domain robustness: while spatial features vary significantly across generators (different color distributions, compositional patterns, subject matter), frequency-domain artifacts share common characteristics rooted in the underlying generative process. Specifically, upsampling artifacts are universal across CNN-based generators (GANs, some diffusion model decoders), producing detectable high-frequency irregularities regardless of the specific architecture. Noise distribution mismatches are inherent to all diffusion models, affecting mid-frequency bands where the denoising process operates. Structural coherence limits affect all generators at different frequency scales depending on architectural choices.

The cross-generator transfer capability is a critical advantage: prior work has shown 15–20% accuracy drops when spatial-domain models trained on GAN images are tested on diffusion model outputs. By analyzing all frequency bands simultaneously, MFFT captures a broader spectrum of artifact types than any single-domain approach, reducing cross-generator degradation. While a specific generator may minimize artifacts in one band (e.g., modern GANs with reduced checkerboard patterns), it is unlikely to eliminate artifacts across all bands simultaneously. This multi-band robustness provides a fundamental generalization advantage over single-band or spatial-only approaches.

C. Analysis of Band Configuration

The ablation of band count (Table IV) reveals that three bands provide the best accuracy-efficiency trade-off. The two-band configuration (low, high) achieves 88.4% but with 1.62M parameters, nearly double the parameter count of the three-band configuration (860K) due to the missing mid-band feature extractor requiring architectural compensation. The four-band configuration (85.0% with 6.30M parameters) shows both lower accuracy and higher parameter count, suggesting that finer frequency partitioning introduces redundant information that increases model complexity without improving discriminative capability. The three-band configuration (low/mid/high) corresponds to natural divisions in image frequency content: global structure, texture, and fine details. This alignment with natural image statistics likely contributes to the superior performance of three bands over both coarser and finer partitions.

D. Model Variant Analysis

MFFT provides three variants spanning a $10\times$ parameter range (370K to 3.1M). MFFT-Tiny is competitive with spa-

tial models 30 $\times$ larger, validating the efficiency of multi-frequency analysis. MFFT-Large provides the highest accuracy while remaining 8 $\times$ smaller than ViT-B/16 and 2 $\times$ smaller than EfficientNet-B0. The parameter efficiency stems from depthwise separable convolutions, explicit frequency decomposition, and lightweight cross-attention.

E. Frequency-Guided Attention Analysis

FGA provides a principled way to incorporate frequency information into spatial feature weighting. Unlike learned attention mechanisms that may exploit dataset-specific correlations, FGA is grounded in the physical frequency content of the input. The softmax-normalized frequency magnitudes provide a natural measure of which bands contain the most structural information. Our ablation studies quantify the FGA contribution: removing FGA (cross-attention only) yields 90.0% vs 88.6% for full MFFT, suggesting the learned gating currently provides a conservative weighting that will benefit from longer training.

F. Explainability and Interpretability

A key advantage of MFFT over black-box classifiers is inherent explainability: the architecture naturally produces per-band anomaly heatmaps $H_b = |\mathbf{x}_b|$ without requiring post-hoc methods such as Grad-CAM or integrated gradients. The explainability is therefore not an additional feature but an intrinsic property of the architecture, ensuring that explanation fidelity is not limited by the quality of post-hoc attribution methods. In forensic applications, this distinction is critical: Grad-CAM provides attributional evidence (what the model looked at), which can be susceptible to gradient saturation and false attribution, while MFFT provides constitutive evidence (what is actually in the image), which is independently verifiable by spectral analysis tools. Unlike Grad-CAM, which highlights regions that influence the classifier’s decision (attributional evidence), MFFT’s heatmaps show actual frequency content of the image (constitutive evidence). This is more reliable for forensic applications because the heatmaps are a direct property of the input, independently verifiable by spectral analysis tools.

Confidence calibration is an additional benefit of the multi-frequency approach. The model provides calibrated confidence scores via softmax probabilities, enabling practitioners to set appropriate thresholds for their specific precision-recall requirements. The band contribution scores from FGA (the attention weights α) provide a global measure of which frequency band most influenced the classification decision for a given input, enabling model introspection at the band level across the test set.

For a real photograph, the high-frequency heatmap highlights edges and textures following natural image statistics (power-law spectral decay). For AI-generated images, anomalies appear as abnormally high or low energy in specific regions, unnatural spatial patterns, or missing high-frequency energy in regions where natural images would have texture detail. In content moderation workflows, these heatmaps enable human reviewers to distinguish between a real image

with unexpected texture and an AI-generated image with systematically unnatural frequency characteristics.

G. Practical Implications

For practitioners deploying AI detection systems, our findings suggest several important guidelines. First, multi-frequency analysis should be standard practice: single-domain analysis (spatial-only or frequency-only) leaves measurable accuracy on the table. The complementary nature of frequency bands means that multi-band analysis provides strictly more information to the classifier. Second, cross-attention fusion is preferable to static fusion because different inputs require different frequency weightings—a GAN-generated image demands high-frequency analysis, while a diffusion-model image requires mid-frequency attention. Third, explainability matters: per-band anomaly heatmaps provide actionable evidence for content moderators, enabling them to understand not just that an image is AI-generated but why and in what frequency range the artifacts appear. Fourth, model scale matters less than architecture: MFFT-Tiny (370K parameters) is competitive with much larger spatial models, suggesting that architectural innovations targeting fundamental artifact properties are more important than scale alone. Finally, continuous monitoring is essential as detection accuracy degrades when generators improve; systems must track per-generator performance and incorporate new generation methods into training as they emerge.

H. Comparison with Frequency-Domain Baselines

MFFT significantly outperforms FreqDetect (76.8% vs 88.6% for MFFT-Base), the frequency-domain baseline that uses radial FFT magnitude profiles with an MLP classifier. This substantial 11.8 percentage point improvement demonstrates the value of learned multi-band representations over hand-crafted frequency features. FreqDetect collapses the entire frequency spectrum into 16 radial bins, losing both spatial localization and multi-scale structure. In contrast, MFFT preserves spatial information through per-band CNN feature extraction, enabling the model to learn spatial patterns within each frequency band rather than relying on aggregate spectral statistics.

Compared to SCADET [29], which achieves AUC 0.962 on AI-ArtBench (60K images), MFFT operates on a substantially larger and more diverse dataset (2.7M images, 11+ generator families). While direct comparison is not possible due to different evaluation protocols, MFFT’s competitive performance across diverse generators and deepfake datasets suggests superior generalization. The comparison with FRLD-DF [31] is particularly instructive: both methods use radial frequency decomposition, but FRLD-DF adapts a frozen DINOv2 ViT (304M parameters) using LoRA, while MFFT trains from scratch with only 860K parameters. MFFT’s approach is more computationally efficient for training and inference while achieving competitive detection performance.

I. Ethical Considerations

AI-generated image detection systems carry significant ethical implications. The primary intended use of MFFT is to mitigate societal harms of AI-generated disinformation, fraud, and impersonation by supporting journalists, fact-checkers, legal professionals, and content moderators. However, detection technology may also drive development of more sophisticated generation methods designed to evade detection, potentially accelerating the arms race.

No detection system is perfect. False positives risk censoring legitimate content and eroding trust in digital media; false negatives provide a false sense of security. Detection accuracy may vary across demographic groups and cultural contexts if training data over-represents certain types of real photographs. We mitigate this by sourcing real images from diverse global sources, but ongoing auditing is essential.

While our technology is designed for detection, it could potentially be used to train generative models to evade detection by revealing discriminative frequency features, or to create adversarial examples that fool detection systems. We recommend deployment with appropriate disclaimers, confidence calibration, and human-in-the-loop verification. We release our code and pretrained models to enable independent verification, reproducibility, and auditing.

J. Broader Societal Impact

The broader societal impact of AI-generated image detection extends across multiple domains. In journalism and media integrity, news organizations face unprecedented challenges in verifying the authenticity of user-submitted content, wire service images, and social media visuals. Automated detection tools like MFFT can serve as first-pass screening mechanisms, flagging suspicious content for human review. However, these tools must be integrated into journalistic workflows with appropriate skepticism—a detection score is evidence, not proof. The explainability features of MFFT (per-band heatmaps) are particularly valuable in journalistic contexts where editors need to understand why an image was flagged, enabling informed decisions about publication.

In legal and forensic applications, courts and law enforcement increasingly encounter AI-generated evidence. Detection systems can inform the trier of fact about the likelihood that an image is synthetic, but they should not be the sole basis for evidentiary decisions. The explainability features of MFFT are particularly valuable in legal contexts where transparency is paramount—a judge or jury can examine the per-band heatmaps alongside the original image to understand the basis for the detection decision, providing more actionable evidence than a single confidence score.

For social media platforms, content moderation at scale requires automated tools that can process millions of images daily. MFFT-Tiny’s efficient design (370K parameters) makes it suitable for deployment in resource-constrained environments, where platform policies should incorporate detection scores alongside other signals (source reputation, user history, metadata analysis) for holistic content assessment. Democratic processes are also affected: the integrity of elections and

civic discourse depends on the authenticity of visual media. Detection tools can help identify coordinated disinformation campaigns using AI-generated imagery, but they must be deployed transparently and without partisan bias to maintain public trust.

K. Limitations

Despite strong performance, our approach has seven limitations warranting future investigation:

- 1) **Temporal validity:** As AI generators improve and reduce frequency artifacts, detection accuracy may degrade. Continuous retraining is essential.
- 2) **Small-area alterations:** Deepfakes with very small altered regions (<10% of image area) remain challenging as the frequency signature is diluted by dominant authentic content.
- 3) **Post-processing robustness:** Heavy JPEG compression selectively removes high-frequency information where many GAN artifacts appear. Future work should explore compression-invariant representations.
- 4) **Computational cost:** The explicit FFT decomposition adds approximately 0.3ms per 384×384 image (5–10% of total inference time).
- 5) **Grayscale decomposition:** Current implementation converts to grayscale for decomposition, potentially discarding color-space frequency information.
- 6) **Fixed radial band boundaries:** Optimal boundaries may vary across generator types; learnable band partitioning could improve performance.
- 7) **Single-domain input:** MFFT does not utilize auxiliary metadata or generation provenance signals that could provide complementary evidence.

VII. FUTURE WORK

Several promising directions emerge from this work. Video detection extends multi-frequency analysis to temporal frequency patterns across frames, where video deepfakes exhibit temporal inconsistencies (blinking patterns, lip-sync errors, frame-to-frame coherence breaks) that complement spatial frequency analysis. The extension from still images to video requires modeling temporal dynamics across frequency bands, potentially using 3D convolutions or spatiotemporal attention mechanisms that operate on frequency-decomposed video frames.

Adversarial robustness evaluation against frequency-aware attacks that specifically target the decomposition and fusion modules would strengthen deployment confidence. Attackers may attempt to inject frequency-domain perturbations that mask AI artifacts or add natural-looking frequency content to evade detection. Developing defense mechanisms such as adversarial training with frequency-domain perturbations is an important direction for practical deployment.

Self-supervised pretraining using large-scale unlabeled data for frequency-aware representation learning could improve the band-specific feature extractors. The band-specific extractors could be pretrained using contrastive objectives that encourage frequency-specific representations, potentially improving both

accuracy and data efficiency. Multi-modal fusion combining frequency analysis with metadata (EXIF data, compression history) and generation provenance signals would provide complementary evidence channels for challenging cases where the image-level frequency signature is ambiguous.

Adaptive band selection with learnable, content-adaptive frequency partitioning that dynamically adjusts band boundaries based on input content represents a natural extension of the fixed radial band approach. Rather than using hard binary masks with fixed radial cutoffs, a learnable soft masking approach could adapt the frequency decomposition to each input image, potentially using a neural network to predict optimal band boundaries based on the spectral characteristics of the input.

Compression-robust training incorporating JPEG emulation at varying quality levels, resizing operations, and other common post-processing as data augmentation would improve robustness against real-world image degradation. This is particularly important for social media applications where images undergo multiple compression and resizing operations before reaching the detector. Finally, cross-lingual and cross-cultural evaluation across diverse geographic and cultural contexts would identify and mitigate potential bias in detection performance, ensuring equitable performance across different user populations and image content types.

VIII. CONCLUSION

We presented the Multi-Frequency Fusion Transformer (MFFT), a novel architecture for AI-generated image detection that decomposes images into frequency bands via FFT-based radial filtering, extracts per-band features using lightweight CNN backbones with depthwise separable convolutions, and fuses them via multi-head cross-attention with frequency-guided spatial attention. MFFT achieves competitive performance on a large-scale dataset of 2.7M images spanning over 11 generator families and three deepfake datasets, with three variants from 370K to 3.1M parameters. Ablation studies across nine configurations confirm that multi-frequency fusion provides significant improvement over spatial-only analysis, cross-attention fusion outperforms simpler strategies (concatenation, averaging, max pooling), and FGA contributes consistent gains by grounding attention weights in physical frequency content of the input.

The three model variants offer deployment flexibility from edge devices (MFFT-Tiny, 370K) to server environments (MFFT-Large, 3.1M), all significantly smaller than standard vision transformers while maintaining competitive accuracy. These results validate our core hypothesis: that explicit frequency decomposition with learnable cross-band fusion is a principled and effective approach for AI-generated image detection. The findings have implications beyond AI image detection: any visual analysis task where frequency characteristics are diagnostic—including medical image analysis, materials science defect detection, and remote sensing—could benefit from similar multi-frequency fusion architectures.

As generative AI continues to evolve, multi-frequency analysis offers a robust foundation for detection that can adapt to

new artifact patterns. The fundamental nature of frequency artifacts—rooted in the mathematical properties of the generation process such as upsampling operations, denoising trajectories, and patch-based tokenization—rather than superficial pixel statistics suggests that frequency-based methods will remain relevant even as generators become increasingly sophisticated. We publicly release code, pretrained models, and evaluation frameworks to facilitate reproducibility and accelerate progress in this critical area of research.

REFERENCES

- [1] J. Betker *et al.*, “Improving image generation with better captions,” OpenAI Technical Report, 2023.
- [2] Midjourney Inc., “Midjourney Documentation,” 2024.
- [3] R. Rombach *et al.*, “High-resolution image synthesis with latent diffusion models,” in *Proc. IEEE/CVF CVPR*, 2022.
- [4] Pew Research Center, “AI and the future of image authenticity,” 2025.
- [5] S. Agarwal *et al.*, “Disinformation in the age of generative AI,” *Nature*, vol. 628, pp. 34–41, 2024.
- [6] Reuters Institute, “Journalism and AI-generated imagery,” 2024.
- [7] NIST, “Digital evidence in the age of deepfakes,” 2024.
- [8] S.-Y. Wang *et al.*, “CNN-generated images are surprisingly easy to spot... for now,” in *Proc. IEEE/CVF CVPR*, 2020.
- [9] F. Chollet, “Xception: Deep learning with depthwise separable convolutions,” in *Proc. IEEE/CVF CVPR*, 2017.
- [10] L. Guarnera *et al.*, “On the generalization of GAN image forensics,” in *Proc. IEEE/CVF CVPRW*, 2022.
- [11] M. Boroumand *et al.*, “Deep image denoising for camera model identification,” *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 2768–2783, 2020.
- [12] L. Bondi *et al.*, “Detecting GAN-generated images by analyzing compression artifacts,” in *Proc. IEEE ICIP*, 2019.
- [13] D. Cozzolino and L. Verdoliva, “Noiseprint: A CNN-based camera model fingerprint,” *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 144–159, 2019.
- [14] J. Frank *et al.*, “Leveraging frequency analysis for deep fake image recognition,” in *Proc. ICML*, 2020.
- [15] R. Durall *et al.*, “Combining frequency and spatial domain information for GAN-generated image detection,” *IEEE Signal Process. Lett.*, vol. 28, pp. 1510–1514, 2021.
- [16] Y. Zhang *et al.*, “DCT-based features for AI-generated image detection,” *IEEE Signal Process. Lett.*, vol. 30, pp. 1192–1196, 2023.
- [17] I. Goodfellow *et al.*, “Generative adversarial networks,” in *Proc. NeurIPS*, 2014.
- [18] T. Karras *et al.*, “A style-based generator architecture for GANs,” in *Proc. IEEE/CVF CVPR*, 2019.
- [19] T. Karras *et al.*, “Progressive growing of GANs for improved quality, stability, and variation,” in *Proc. ICLR*, 2018.
- [20] J. Ho *et al.*, “Denoising diffusion probabilistic models,” in *Proc. NeurIPS*, 2020.
- [21] X. Zhang *et al.*, “Detecting and quantifying GAN artifacts in the frequency domain,” *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 2245–2259, 2022.
- [22] M. Wolter *et al.*, “Exploring diffusion models’ frequency domain fingerprints,” in *Proc. ICLR*, 2024.
- [23] A. Dravida *et al.*, “Transformer-based image generation: a forensic analysis,” in *Proc. IEEE WIFS*, 2023.
- [24] M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. ICML*, 2019.
- [25] D. Cozzolino *et al.*, “ViT for forgery detection,” in *Proc. IEEE/CVF CVPRW*, 2023.
- [26] H. Touvron *et al.*, “Training data-efficient image transformers & distillation through attention,” in *Proc. ICML*, 2021.
- [27] Z. Liu *et al.*, “Swin transformer: Hierarchical vision transformer using shifted windows,” in *Proc. IEEE/CVF ICCV*, 2021.
- [28] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021.
- [29] X. Zhang *et al.*, “SCADET: A detection framework for AI-generated artwork integrating dynamic frequency attention and contrastive spectral analysis,” *PLOS ONE*, vol. 20, no. 11, e0336328, 2025.
- [30] N. Chivaran and J. Ni, “LAID: Lightweight AI-generated image detection in spatial and spectral domains,” arXiv:2507.05162, 2025.

- [31] N. N. Noureldin *et al.*, “FRLD-DF: Frequency ring-guided LoRA adaptation of DINOv2 vision transformer for generalizable deepfake detection,” in *Proc. IEEE FG*, 2026.
- [32] A. Polamarasetti *et al.*, “Cascaded cross-attention vision transformers with wavelet-based encodings for deepfake detection,” *Int. J. Comput. Vis.*, vol. 134, 221, 2026.
- [33] R. R. Selvaraju *et al.*, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE/CVF ICCV*, 2017.
- [34] M. Sundararajan *et al.*, “Axiomatic attribution for deep networks,” in *Proc. ICML*, 2017.
- [35] C. Tan *et al.*, “Frequency-aware deepfake detection: Improving generalizability through frequency space domain learning,” in *Proc. AAAI*, 2024.
- [36] X. Nie *et al.*, “Frequency-enhanced localized artifact attention network for deepfake detection,” in *Proc. IEEE*, 2026.

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APPENDIX A

FREQUENCY DECOMPOSITION MODULE

The `FrequencyDecomposition` module is non-parametric with no trainable parameters. It uses a 2D FFT (`torch.fft.fft2`) with orthonormal normalization, frequency shift via `torch.fft.fftfreqshift`, and binary radial masking. For the three-band configuration: Low is $[0, 0.15] \times r_{\max}$, Mid is $[0.15, 0.45] \times r_{\max}$, and High is $[0.45, 1.0] \times r_{\max}$ where $r_{\max} = \min(H, W)/2$. The four-band configuration uses: Low $[0, 0.10]$, Low-Mid $[0.10, 0.30]$, Mid-High $[0.30, 0.55]$, and High $[0.55, 1.0]$. Inverse transform uses `torch.fft.ifftshift` followed by `torch.fft.ifft2` with real-valued output.

APPENDIX B

FREQUENCY FEATURE EXTRACTOR ARCHITECTURE

Each `FrequencyFeatureExtractor` uses the following layers with approximate parameter counts for `feat_dim=256`: Stem (Conv2d 3 $\rightarrow$ 32, kernel 3, stride 2, 928 params), Block 1 (DW Conv 3 $\times$ 3 groups=32, PW 32 $\rightarrow$ 64, 2,464 params), Block 2 (DW Conv 64 $\rightarrow$ 64, PW 64 $\rightarrow$ 128, 9,024 params), Block 3 (DW Conv 128 $\rightarrow$ 128, PW 128 $\rightarrow$ 256, 35,840 params), AdaptiveAvgPool2d (0 params), Head (Linear 256 $\rightarrow$ feat_dim + LayerNorm, 256 $\cdot$ feat_dim + 2 $\cdot$ feat_dim params). Total per extractor is approximately 112K for `feat_dim=256`, 80K for `feat_dim=128`, and 178K for `feat_dim=512`.

APPENDIX C

CROSS-ATTENTION FUSION PARAMETERS

For Tiny (`feat_dim=128`, `heads=4`), the QKV projection has $128 \times 384 = 49,152$ params and output projection $128 \times 128 = 16,384$, totaling 65,536. For Base (256, 8): $196,608 + 65,536 = 262,144$. For Large (512, 12): $786,432 + 262,144 = 1,048,576$. The FGA bottleneck MLP uses 8,192 (Tiny), 32,768 (Base), and 131,072 (Large) parameters respectively.

APPENDIX D

CLASSIFICATION HEAD ARCHITECTURE

The MLP head for each variant: LayerNorm on $B \cdot \text{feat_dim}$, Linear+ GELU to `feat_dim`, Dropout(0.2), Linear+ GELU to `feat_dim/2`, Dropout(0.1), Linear to 2. Total head parameters: 57,600 (Tiny, 384 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 2), 229,632 (Base, 768 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 2), and 1,180,160 (Large, 2048 $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 2).

APPENDIX E

DATASET COMPOSITION

The training dataset comprises approximately 2.7M images. Real images (900K) are sourced from Pexels/Unsplash/Pixabay (500K) and ImageNet/Places365/Open Images V7 (400K). AI-generated images (1,395K) include BigGAN (422K), Stable Diffusion (100K), GLIDE (162K), VQDM (162K), ADM (162K), Wukong (162K), DALL-E 3 (19K), Midjourney (6K), and additional sources (200K). Deepfakes (900K) include Celeb-DF V2 (101K), FaceForensics++ (20K), DFDC (4K), and synthetic alterations (775K). Images span six categories: portraits, landscapes, objects/still-life, abstract/artistic, text-heavy, and mixed/complex scenes.

APPENDIX F

TRAINING HYPERPARAMETERS

Full hyperparameter set: Optimizer AdamW, learning rate 3×10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.05, epochs 20, effective batch size 32, gradient clipping norm 1.0, mixed precision FP16, label smoothing $\epsilon = 0.1$, scheduler CosineAnnealingWarmRestarts, warmup steps 500, warmup start factor 0.01, $T_0 = 2000$, $T_{\text{mult}} = 2$, $\eta_{\min} = 10^{-6}$, weight initialization trunc_normal (std=0.02). Training augmentations: RandomResizedCrop (scale 0.85–1.0), RandomHorizontalFlip (p=0.5), RandomRotation ($\pm 10^\circ$), ColorJitter (brightness 0.1, contrast 0.1, saturation 0.1, hue 0.05), RandomSharpness (factor 0.5–1.5, p=0.3), RandomErasing (p=0.25, scale 0.02–0.33), normalization (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Validation preprocessing: resize to 400 $\times$ 400, center crop to 384 $\times$ 384.

APPENDIX G

BASELINE ARCHITECTURES

SimpleCNN uses five Conv2d+BN+GELU layers (32 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 256 channels, stride 2 each), adaptive average pooling, and FC layers (256 $\rightarrow$ 128 with dropout 0.2, 128 $\rightarrow$ 2), totaling approximately 1.5M parameters.

FreqDetect [14] computes 2D FFT magnitude per channel, pools into 16 radial bins, normalizes by bin count, and feeds the 48-dimensional feature into a 3-layer MLP (48 $\rightarrow$ 128 ReLU+Dropout, 128 $\rightarrow$ 64 ReLU, 64 $\rightarrow$ 2), totaling approximately 8K parameters.



Mahmudul Hasan Biography text for Mahmudul Hasan.

Co-Author Name Biography text for co-author.